



**L9 (CONT.)**  
**FOOD COMPOSITION TABLE**  
**NUTRIENT DENSITY**

# **FOOD COMPOSITION TABLE**

- **Food composition table giving the approximate nutrient content of foods are a necessary tool for evaluating diets .**
- **Tables are expressed in either 100 g portions of a food, the edible portion of ( 1 lb) of food or portion commonly consumed.**
- **Energy is expressed as calories and nutrients are generally indicated using metric measure or international units ( retinol equivalents for vitamin A )**

# **EFFECT ON FOOD COMPOSITION TABLE**

- 1- Growing conditions ( sunlight, soil, climate ) affect the nutrient content of the same variety of a food .**
- 2- There is variability in a plant food depending upon maturity , variety , season and plant part**
- 3- Storage, processing and preparation influence nutrient values .**
- 4- The total amount of the nutrient listed may not be biologically available to an individual .**

# FOOD COMPOSITION TABLE

| NO | Food          |           | Water<br>g | Protein<br>g | Fat<br>g | Fiber<br>g | CHO<br>g | energy | Vit A | B1<br>mg | B2<br>mg | Vit C<br>mg | Na<br>mg | K<br>mg | Ca<br>mg | Mg<br>mg | Fe<br>mg | ZN<br>mg | Cu<br>mg | P<br>mg |
|----|---------------|-----------|------------|--------------|----------|------------|----------|--------|-------|----------|----------|-------------|----------|---------|----------|----------|----------|----------|----------|---------|
| 1  | Rice          | ارز       | 11.2       | 7.4          | 0.6      | 0.5        | 80.1     | 355    | 0     | 0.06     | 0.04     | 0           | 11       | 127     | 22       | 30       | 0.8      | 1.2      | 0.08     | 108     |
| 2  | Sorghum       | ذرة بيضاء | 10.5       | 8.8          | 3.6      | 2.2        | 75.3     | 369    | 0     | 0.18     | 0.05     | 0           | 10       | 385     | 30       | 134      | 5.0      | 4.2      | 0.08     | 250     |
| 3  | Wheat         | قمح       | 12.4       | 11.8         | 1.8      | 2.4        | 72.1     | 352    | 284   | 0.61     | 0.12     | 0           | 37       | 315     | 35       | 118      | 2.8      | 3.8      | 0.36     | 412     |
| 4  | egg           | بيض       | 75.2       | 12.6         | 10.8     | 0          | 0.3      | 149    | 918   | 0.14     | 0.36     | 0           | 155      | 174     | 62       | 15       | 2.5      | 1.5      | 0.14     | 218     |
| 5  | Cow milk      | حليب      | 88.2       | 3.5          | 3.5      | 0          | 4.1      | 62     | 165   | 0.03     | 0.17     | 0           | 62       | 152     | 122      | 14       | 0.06     | 0.40     | 0.01     | 104     |
| 6  | Carrot        | جزر       | 89.4       | 1.2          | 0.2      | 1.0        | 8.4      | 40     | 1025  | 0.05     | 0.04     | 8           | 62       | 345     | 40       | 18       | 1.1      | 0.42     | 0.02     | 48      |
| 7  | Garden rocket | جرجير     | 89.9       | 2.5          | 0.5      | 1.0        | 5.4      | 36     | 1200  | 0.14     | 0.21     | 140         | 33       | 257     | 127      | 13       | 8.3      | 1.8      | 0.26     | 40      |
| 8  | Jaw's mallow  | ملوخية    | 83.9       | 5.0          | 1.0      | 1.5        | 7.8      | 60     | 4050  | 0.50     | 0.32     | 80          | 180      | 32      | 370      | 17       | 5.6      | 0.13     | 0.04     | 55      |
| 9  | Parsley       | بقدونس    | 84.6       | 3.3          | 0.4      | 1.3        | 9.5      | 55     | 4200  | 0.15     | 0.32     | 156         | 30       | 723     | 210      | 39       | 5.1      | 1.0      | 0.18     | 54      |
| 10 | Beef          | لحم بقرى  | 67.4       | 19.6         | 11.9     | 0          | 0        | 186    | 12    | 0.07     | 0.11     | 0           | 72       | 375     | 15       | 22       | 3.4      | 4.8      | 0.10     | 174     |
| 11 | Lamb          | لحم ضانى  | 58.0       | 16.5         | 10.3     | 0          | 0        | 287    | 30    | 0.12     | 0.18     | 0           | 65       | 300     | 15       | 15       | 2.5      | 2.7      | 0.6      | 163     |
| 12 | liver         | كبد عجل   | 70.5       | 18.5         | 4.0      | 0          | 5.4      | 132    | 40000 | 0.26     | 2.7      | 25          | 80       | 302     | 10       | 14       | 7.0      | 4.1      | 2.2      | 286     |
| 13 | orange        | برتقال    | 85.5       | 1.1          | 0.3      | 0.6        | 12.7     | 52     | 215   | 0.08     | 0.04     | 55          | 3        | 181     | 37       | 7        | 0.3      | 0.13     | 0.03     | 28      |
| 14 | Apricot       | شمش       | 84.6       | 0.9          | 0.4      | 0.7        | 13.4     | 61     | 2550  | 0.04     | 0.06     | 12          | 3        | 460     | 15       | 11       | 0.5      | 0.05     | 0.10     | 25      |
| 15 | Banana        | موز       | 75.2       | 1.3          | 0.3      | 0.6        | 22.3     | 97     | 160   | 0.04     | 0.05     | 8           | 3        | 350     | 10       | 30       | 0.6      | 0.12     | 0.18     | 25      |
| 16 | Dates dried   | تمر       | 23.0       | 2.3          | 0.5      | 2.2        | 72.5     | 304    | 36    | 0.08     | 0.1      | 0           | 0        | 690     | 62       | 44       | 2.9      | 0.4      | 0.3      | 61      |
| 17 | Mango         | مانجو     | 81.8       | 0.8          | 0.3      | 1.1        | 16.6     | 65     | 3750  | 0.04     | 0.04     | 30          | 10       | 200     | 15       | 8        | 0.5      | 0.18     | 0.11     | 15      |
| 18 | Almonds       | لوز       | 4.6        | 17.6         | 55.8     | 2.5        | 19.3     | 650    | 70    | 0.26     | 0.72     | -           | 7.0      | 680     | 215      | 230      | 4.5      | 3.2      | 0.13     | 385     |

# **Dietary guidelines for Americans**

## **Purpose of Guidelines**

- 1- Good nutrition and good health go hand in hand.**
- 2- Proper nutrition is essential for the healthy growth and development of kids and adolescents.**
- 3-The major causes of illness and death in the United States are related to poor diet and an inactive lifestyle.**
- 4- Especially true with the high rates of obesity in our country - obesity is a result of an energy imbalance from consuming more calories than are burned.**

# **Dietary guidelines for Americans**

## **Messages of the Dietary Guidelines for Americans, 2010:**

### **1-Balancing Calories**

**Balance the calories you consume from foods and beverages with the calories you use each day. Find out how many calories you need for a day as a first step to managing your weight.**

**You can balance calories by :**

## **A-Enjoy your food, but eat less.**

**Take the time to fully enjoy your food as you eat it.**

**Eating too fast or when your attention is elsewhere may lead to eating too many calories.**

## **B-Avoid oversized portions.**

**Use a smaller plate, bowl and glass. When eating out, choose a smaller size option, share a dish, or take home part of your meal.**

## **2-Foods to Increase**

**Eat more vegetables, fruits, whole grains, and fat-free or 1% milk and dairy products. These foods have the nutrients you need for health – including potassium, calcium, vitamin D and fiber. Make them the basis for meals and snacks.**



## **A-Make half your plate fruits and vegetables.**

Choose red, orange and dark-green vegetables like tomatoes, sweet potatoes, and broccoli, along with other vegetables for your meals. Add fruit to meals as part of main or side dishes or as dessert.

## **B-Make at least half your grains whole grains.**

To eat more whole grains, substitute a whole-grain product for a refined product – such as eating whole-wheat bread instead of white bread or brown rice instead of white rice.

## **C-Vary your protein foods.**

**Choose a variety of protein foods, which include seafood, lean meat and poultry, eggs, beans and peas, soy products and unsalted nuts and seeds.**

## **D-Switch to fat-free or low-fat (1%) milk.**

**They have the same amount of calcium and other essential nutrients as whole milk, but fewer calories and less saturated fat.**

## **3-Foods to Reduce**

**Cut back on foods high in solid fats, added sugars, and salt. They include cakes, cookies, ice cream, candies, sweetened drinks, pizza, and fatty meats like ribs, sausages, bacon and hot dogs. Use these foods as occasional treats, not every day foods.**

## **A: Compare sodium in foods.**

**Use the Nutrition Facts label to choose lower sodium versions of foods like soup, bread, and frozen meals. Select canned foods labeled, “low sodium”, “reduced sodium” or “no added salt.”**

## **B: Drink water instead of sugary drinks.**

**Cut calories by drinking water or unsweetened beverages. Soda, energy drinks, and sports drinks are a major source of added sugar, and calories.**

# NUTRIENT DENSITY

## **Definition**

**Nutrient density the ration between the calorie content and the nutrient composition in food, is a new system of indicating food quality.**

## **The benefit of nutrient density:**

**The resulting scores are useful in .**

- 1- helping health professionals**
- 2- consumers assess the nutritional quality of individual foods and diets .**

## **What does calorie dense mean?**

- If a food is calorie dense, that usually means it is high in energy and low in nutrients. This means it is an EMPTY food. They are nutritionally poor food choices and contain much more calories than nutrients like vitamins and minerals.
- If the body does not use all of the energy that was consumed, it gets stored in the body as fat which could lead to weight gain, poor heart health, and onset of diabetes.
- These are typically highly processed foods which mean they are stripped of their nutrients and have added fat, sugar, and salt.



## **HOW DO I CHOOSE NUTRIENT DENSE FOODS:**

**1-Emphasize Quality** – Choose more nutrient dense varieties of products when available.

**2-Encourage Diversity** – Eat foods from at least 3 food groups at meals, and 2 at snacks.

**3-Implement Gradually** – Ease the household into healthier options, one or two at a time.

**4-Be Strategic** – Separate the changes someone will and will not make.

**5-Foster Creativity** – Make healthy options fun and exciting.

# NUTRIENT DENSITY

Low Density

High Density

| Capri Sun (16 oz.) | Nutrients   | Orange Juice (16 oz) |
|--------------------|-------------|----------------------|
| 220                | Calories    | 220                  |
|                    | Total Fat   |                      |
| 60 grams           | Sugars      | 48 grams             |
|                    | Protein     | 4 grams              |
|                    | Potassium   | 900 mg (26%)         |
|                    | Vitamin C   | 120 mcg (200%)       |
|                    | Vitamin A   | 728 IU (18%)         |
|                    | Folate      | 120 mcg (30%)        |
|                    | Calcium     | 52 mg (4%)           |
|                    | Magnesium   | 54 mg (12%)          |
|                    | Thiamin     | 0.5 mg (20%)         |
|                    | Vitamin B-6 | 0.3 mg (8%)          |



# Nutrient Density

These beverages have about the same number of calories in a 12-ounce serving, but the cola and tea provide few nutrients besides added sugars (about 10 teaspoons). Orange juice and milk are rich in many nutrients. (Color bars represent percentage of recommended daily intake or limit for each nutrient.)

| Nutrient            | Recommended Daily Intake* | Regular Cola<br>  | Bottled Iced Tea<br>  | Orange Juice<br>  | Low-Fat (1%) Milk<br>  |
|---------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calories            | 2000 calories             | 152 calories                                                                                                                                                                        | 150 calories                                                                                                                                                                                | 168 calories                                                                                                                                                                            | 150 calories                                                                                                                                                                                 |
| Carbohydrate        | 300 g                     | 38 g                                                                                                                                                                                | 37.5 g                                                                                                                                                                                      | 40.5 g                                                                                                                                                                                  | 18.0 g                                                                                                                                                                                       |
| <i>Added sugars</i> | 40 g                      | 38 g                                                                                                                                                                                | 34.5 g                                                                                                                                                                                      |                                                                                                                                                                                         |                                                                                                                                                                                              |
| Fat                 | 65 g                      |                                                                                                                                                                                     |                                                                                                                                                                                             |                                                                                                                                                                                         | 3.9 g                                                                                                                                                                                        |
| Protein             | 55 g                      |                                                                                                                                                                                     |                                                                                                                                                                                             |                                                                                                                                                                                         | 12 g                                                                                                                                                                                         |
| Calcium             | 1000 mg                   | 11 mg                                                                                                                                                                               |                                                                                                                                                                                             | 33 mg                                                                                                                                                                                   | 450 mg                                                                                                                                                                                       |
| Potassium           | 2000 mg                   | 4 mg                                                                                                                                                                                |                                                                                                                                                                                             | 710 mg                                                                                                                                                                                  | 570 mg                                                                                                                                                                                       |
| Vitamin A           | 800 µg                    |                                                                                                                                                                                     |                                                                                                                                                                                             | 30 µg                                                                                                                                                                                   | 216 µg                                                                                                                                                                                       |
| Vitamin C           | 75 mg                     |                                                                                                                                                                                     |                                                                                                                                                                                             | 145.5 mg                                                                                                                                                                                | 3.6 mg                                                                                                                                                                                       |
| Vitamin D           | 5 µg                      |                                                                                                                                                                                     |                                                                                                                                                                                             |                                                                                                                                                                                         | 3.7 µg                                                                                                                                                                                       |
| Folate              | 400 µg                    |                                                                                                                                                                                     |                                                                                                                                                                                             | 160 µg                                                                                                                                                                                  | 20 µg                                                                                                                                                                                        |

\*Recommended intakes and limits for a 20-year-old woman consuming 2000 calories per day.

# Comparison of Nutrient Density

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# HIGH NUTRIENT DENSITY FOOD

- ❖ butter, margarine,
- ❖ whipped cream, cream cheese,
- ❖ sour cream, salad dressings,
- ❖ mayonnaise,
- ❖ honey, jam, sugar
- ❖ dried fruits, cottage or ricotta cheese,
- ❖ whole milk, powdered milk,
- ❖ ice cream, yogurt, eggs,
- ❖ nuts, seeds, wheat germ, peanut butter.

# The good, the bad and the really unhealthy

Yale University's Overall Nutritional Quality Index (scores out of 100)

|                               |     |
|-------------------------------|-----|
| Broccoli                      | 100 |
| Blueberries                   | 100 |
| Okra                          | 100 |
| Orange                        | 100 |
| Green Beans                   | 100 |
| Pineapple                     | 99  |
| Radish                        | 99  |
| Summer squash                 | 98  |
| Apple                         | 96  |
| Green cabbage                 | 96  |
| Tomato                        | 96  |
| Clementine                    | 94  |
| Watermelon                    | 94  |
| Mango                         | 93  |
| Red onions                    | 93  |
| Non-fat milk                  | 91  |
| Fresh figs                    | 91  |
| Grapes                        | 91  |
| Banana                        | 91  |
| Avocado                       | 89  |
| Oatmeal                       | 88  |
| Atlantic salmon fillet        | 87  |
| Blackberries                  | 83  |
| Cod fillet                    | 82  |
| Raw almonds                   | 82  |
| Iceberg lettuce               | 82  |
| Raw pecans                    | 82  |
| Rocket                        | 82  |
| Brown rice                    | 82  |
| Snapper                       | 82  |
| Milk (1% fat)                 | 81  |
| Oysters                       | 81  |
| Prawn                         | 75  |
| Couscous                      | 72  |
| Clams                         | 71  |
| Raw pistachios                | 70  |
| Unbuttered, unsalted popcorn  | 69  |
| Canned tuna in oil, drained   | 67  |
| Monkfish fillet               | 64  |
| Vegetarian split-pea soup mix | 63  |
| Instant oatmeal               | 61  |
| Canned pineapple in juice     | 60  |
| White rice                    | 57  |
| Sodium-free sparkling water   | 56  |
| Milk (2% fat)                 | 55  |
| Canned kidney beans           | 53  |
| Milk (whole)                  | 52  |
| Scallops                      | 51  |



|                                   |    |
|-----------------------------------|----|
| Pasta                             | 50 |
| Canned peas                       | 49 |
| Prunes                            | 45 |
| Vanilla yoghurt                   | 43 |
| Orange juice                      | 39 |
| Skinless chicken breasts          | 39 |
| Canned peaches in light Syrup     | 37 |
| Lobster                           | 36 |
| Flank steak (beef)                | 34 |
| Dried apples                      | 34 |
| Tomato juice                      | 32 |
| Condensed split-pea soup with ham | 32 |
| Minced sirloin                    | 30 |
| Chicken drumstick                 | 30 |
| Enriched white bread              | 29 |
| Whole chicken with skin           | 28 |
| Leg of lamb                       | 28 |
| Ham (whole)                       | 27 |
| Raisins                           | 26 |
| Hamburger (75% lean)              | 25 |
| Apple chips                       | 24 |
| Coconut                           | 24 |
| Green olives                      | 24 |
| Pork baby back ribs               | 24 |
| Bagel                             | 23 |
| Condensed tomato soup             | 23 |
| Peanut butter                     | 23 |
| Sherbet                           | 23 |
| Reduced-fat sour cream            | 22 |
| Condensed cream of broccoli soup  | 21 |
| Salted, dry-roasted peanuts       | 21 |
| Instant chocolate pudding         | 20 |
| Fried egg                         | 18 |
| Swiss cheese                      | 17 |
| Diet fizzy drinks                 | 15 |
| Non-streaky bacon                 | 13 |
| Pretzel sticks                    | 11 |
| Dark chocolate                    | 10 |
| White bread                       | 9  |
| Salami                            | 7  |
| Hot dog                           | 5  |
| Cheese puffs                      | 4  |
| Milk chocolate                    | 3  |
| Apple pie                         | 2  |
| Crackers                          | 2  |
| Fizzy drinks                      | 1  |
| Popsicle                          | 1  |

For more food scores visit [nuval.com](http://nuval.com)